

$$u_t + \sigma u_x = 0 \quad 0 \leq x \leq 1 \quad \sigma > 0 \quad (1)$$

$$u(0, x) = \sin(2\pi x) \quad (2)$$

periodic boundary conditions:

$$u(t, x) = u(t, x + 1) \quad (3)$$

Method of lines:

$$i = 0, \dots, I - 1 \quad (4)$$

$$x_i = i\Delta x \quad (5)$$

$$\Delta x = \frac{1.0}{I} \quad (6)$$

Discretize in space, but not in time.

$$u_i(t) \approx u(t, x_i) \quad (7)$$

$$u_i(0) = \sin(2\pi x_i) \quad 0 \leq i \leq I - 1 \quad (8)$$

$$\frac{du_i}{dt} = -\sigma \frac{u_i(t) - u_{i-1}(t)}{\Delta x} \quad 0 < i < I \quad (9)$$

$$\frac{du_0}{dt} = -\sigma \frac{u_0(t) - u_{I-1}(t)}{\Delta x} \quad (10)$$

We solve the above ODE system for $0 \leq t \leq T$.

Quantifying the error:

$$E = \max_{0 \leq i \leq I-1} |u_i(T) - u^{\text{exact}}(T, x_i)| \quad (11)$$